

Data Preprocessing

UW ECE 657A - Core Topic

Mark Crowley

Lecture Outline

- 1 Data Something
 - Types of Data
 - Data Representations
- 2 Preparation of Data
- 3 Data Examination and Cleaning
 - Accuracy, Completeness, Consistency, Interpretability
 - Filling in Missing Data
 - Smoothing with Bins and Windows
 - Normalization and Standardization
 - Scaling
- 4 Data Reduction
 - via Sampling
- 5 Self Study Topics
 - Measuring Similarity
 - Training and Validation Methodology

Announcements

- Time Zone Poll - go online and fill it out
- Pairs, Crews, Houses - more to come tonight
- Detailed Schedule - same

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Data Data Something Something...

- **Data Sources:** measurements from sensors, records, files, document, archives, transactions.
- **Data Modeling:** Creating a structure, organization, function or an abstract view of the data.
- **Data Analysis:** Transforming or operating on data to extract useful information, knowledge or conclusions.
- **Data Mining:** Carrying this further to discover unforeseen or hidden patterns in the data.

Big Data

Big data is about quality and performance given **huge** amounts of data, that is not primarily the focus of this course. But the tools and analysis methods we learn are part of the basis you need to deal with Big Data.

- Volume** - Large amounts of data, social networks, phone, location, embedded systems, environmental, satellites, "full firehose"
- Velocity** - Streaming, online data, arriving quickly, time series, real-time
- Variety** - Heterogeneous (many types), many sources, category data, numerical data, continuous/discrete, text, images, audio, video

Big Data

Some people say it is also includes:

Veracity - Solution requires: Accuracy, Confidence, Precision, Error

Variability - Changes moment to moment, distribution can change at different times (seasonal, trends, fads, memes)

Complexity - Combinatorial connections between entities in the data, form networks, hierarchies

Types of Data

- Nominal
- Ordinal
- Interval
- Ratio

Types of Data (Qualitative)

- **Nominal**: no implication for quantity (qualitative)
e.g occupation: engineer, teacher, dentist, bus driver. Or Color value: blue, red, green,
Binary is a special case 0,1. ($=$, $\neq$,)
- **Ordinal**: relative ranking among values
(i.e. order in relation to each other.
e.g (hi,med,lo), (disagree,neutral, agree),(5,3,1), ($<$, $>$)

Intervals

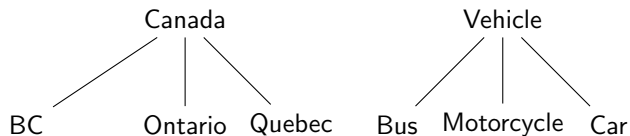
- **Interval:** like the ordinal type but on a scale of equal-size units
i.e. a unit of measurement exists.
- Interpretation of numbers depends on the unit. The interval (range) is important for the interpretation. (+,-)
- E.g the significance of a mark of 10 will be different if the interval is 0-10 from that of interval of 0-100.
- Interval numbers represent differences between values, not absolute quantities.
- Temperature in C or F is an interval attribute

Ratios

- **Ratio:** like the interval in terms of order and uniformity but the scale has an inherent zero-point. ($*$, $/$)
- e.g Kelvin temperature scale for heat. $0K$ means no heat, $50K$ is double the heat of $25K$.
- We cannot say that for the Celsius scale, $0^{\circ}C$ is the amount of heat at freezing point.
- Used for physical quantities: height, weight, length etc.
- Also locations, distance, money

Structural Data

- Values are represented in a tree or hierarchy or graph
- Hierarchical structure could be whole-part, abstract-specific, classes-instances

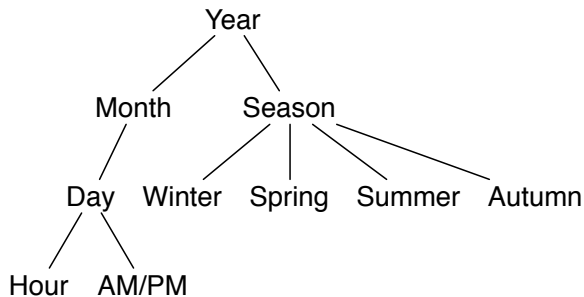


Examples of tree structured data.

Structural Data

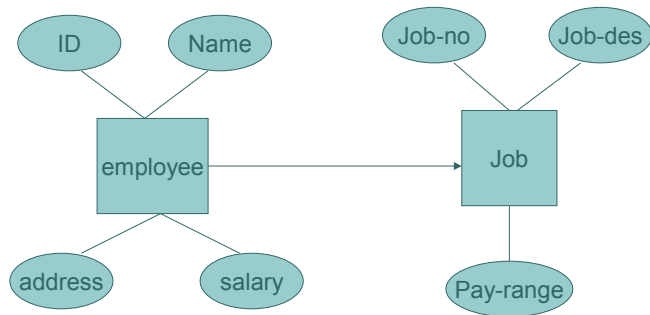
Graphs or Trees

Often we'll instead be presented with data of the kind in Figure ?? which has a mixture of the types of structures and data types. For this reason we must take care to find out as much as possible about the data without assuming a tree structure implies any particular relationship.



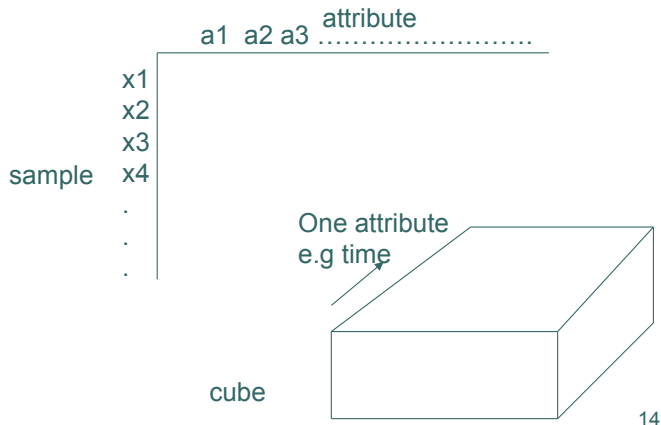
Databases

- data that has the same structure (schema) or abstract view independent of the physical layer



Lists/Vector/Matrix/Data Cube

- mainly table or vector or attribute-sample matrix which provides relational view



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Data Preprocessing Overview

- Examination of Data: Quality
 - accuracy, completeness, consistency, interpretability
- Data Cleaning: missing values, outliers, noise
- Transformation:
 - Smoothing with bins and windows
 - Normalization and Scaling
- Data Reduction: dimensionality, numerosity

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Data examination

Data Quality:

- Accuracy: incorrect (eg.birthdates), inaccurate, transmission errors, duplicates
- Completeness: not recorded values, unavailable, ..
- Consistency: delete inconsistent data? acquire more data? average?
- Interpretability: how easily the data can be understood, correction of errors or removal of inconsistent data could make it harder to interpret

Data Cleaning- Examining the data to correct for

- Missing values: could rows or columns, data points missing (some subpopulation under-represented), or some feature missing (some sensors provided no data during a certain period)
- Outliers: a particular subset of datapoints (usually a small subset) that are abnormal in some way.
- Noise: some stochastic/random/probabilistic deviation from 'true" or "nominal" values that affects all datapoints globally.

Missing Values

Missing some data entirely is never the situation you want to be in. No matter what you do, you will be dealing with an incomplete view of the situation and your solution can only ever be an approximation to the right answer.

One way to reduce this impact is to try to **fill in the missing data** with some *reasonable guess* as to what the value would be:

- Use the mean value of an attribute (or the majority nominal value) to fill in missing values
- If there are classes, use *attribute mean* (or majority nominal value) for all samples in the same class
- Can use *prediction* or *interpolation* to fill in missing values : linear, polynomial, spline
- Can remove the samples that have too many values missing

Example: EPA Data Cleaning Rules

Goal of Substitution Protocols: "The goal of substitution should be to replace missing data with a 'best estimate' so as to minimize the probable error of the estimate. The following suggestions have been prioritized in order of increasing probable error."

- **Persistence** - Persistence is the use of data from the previous time period (hour). This procedure is applicable for most meteorological variables for isolated one-hour gaps; caution should be used when the gaps occur during day/night transition periods.
- **Interpolation** - This procedure is applicable for most meteorological variables for isolated one-hour gaps and, depending on circumstances, may be used for more extended periods (several hours) for selected variables; e.g., temperature. As in the case of persistence, caution should be used when the gaps occur during day/night transition periods.
- **Profiling** - Profiling (profile extrapolation) refers to the procedure in which missing data for one level in a multi-level data base (e.g., data from a meteorological tower) is replaced by an estimate based on data from an alternative level or levels in the same data base.

Example: EPA Data Cleaning Rules

Caveats:

- "Substitution should not be used to attain the 90% completeness requirement for regulatory modeling applications."

Dealing with Outliers or Noise

- **Detection:**

- Use histograms to detect outliers
- Use difference between mean, mode, median to indicate outliers
- Use clustering to detect outliers.
- Observe fluctuation in the values
- Inconsistent values (negative values for positive attributes)

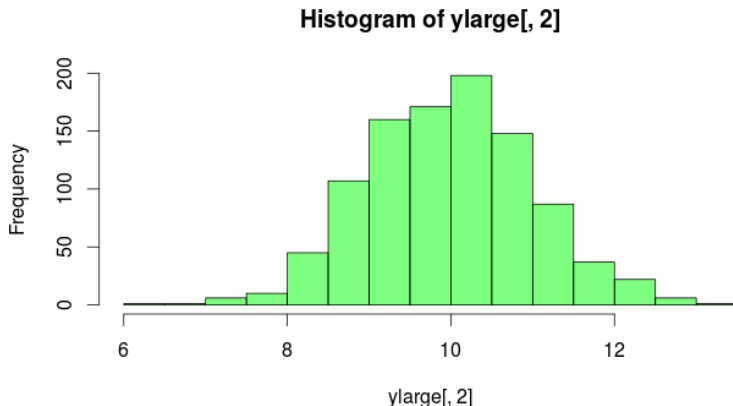
- **Fixing:**

- Remove samples that are way out of range.
- Smoothing the data to get rid of fluctuations.
- Use logic check to correct inconsistency.
- Use prediction methods or fitting.

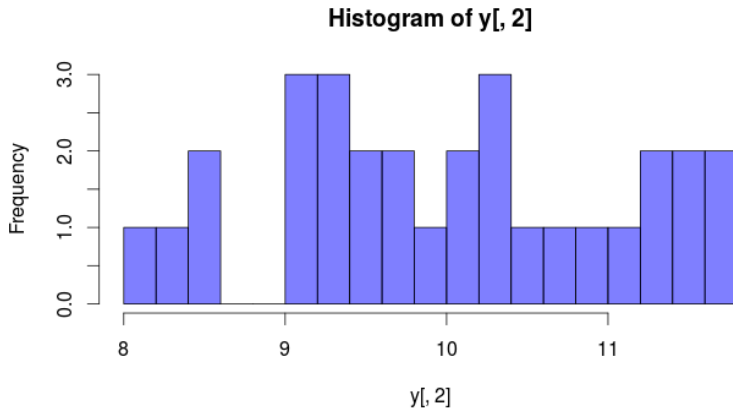
Anomaly Detection: Related, what if the outlier is what you are looking for?

Histograms for Outlier Detection

- Histogram is a bar plot of values vs frequency
- Values divided into **bins** (ranges of values)



Histograms for Outlier Detection



Binning Methods for Smoothing

Data Smoothing: Focus here is not correcting the data but softening it.

- Sort data and partition into bins
 - equal width
 - equal frequency by number of samples
- Smooth the values in each bin by:
 - replacing with the mean or median
 - replacing with the nearest bin boundary value

Binning Example

Sorted data: [4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34]

Using 3 bins of equal 4 samples.

Bin 1	Bin 2	Bin 3
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Binning Example

Sorted data: [4,8,9,15,21,21,24,25,26,28,29,34]

Using 3 bins of equal 4 samples.

	Bin 1	Bin 2	Bin 3
Binned Data:	4,8,9,15	21,21,24,25	26,28,29,34

Binning Example

Sorted data: [4,8,9,15,21,21,24,25,26,28,29,34]

Using 3 bins of equal 4 samples.

	Bin 1	Bin 2	Bin 3
Binned Data:	4,8,9,15	21,21,24,25	26,28,29,34
means	9,9,9,9	23,23,23,23	29,29,29,29

Binning Example

Sorted data: [4,8,9,15,21,21,24,25,26,28,29,34]

Using 3 bins of equal 4 samples.

	Bin 1	Bin 2	Bin 3
Binned Data:	4,8,9,15	21,21,24,25	26,28,29,34
means	9,9,9,9	23,23,23,23	29,29,29,29
boundaries	4,4,4,15	21,21,25,25	26,26,26,34

Smoothing within a Window

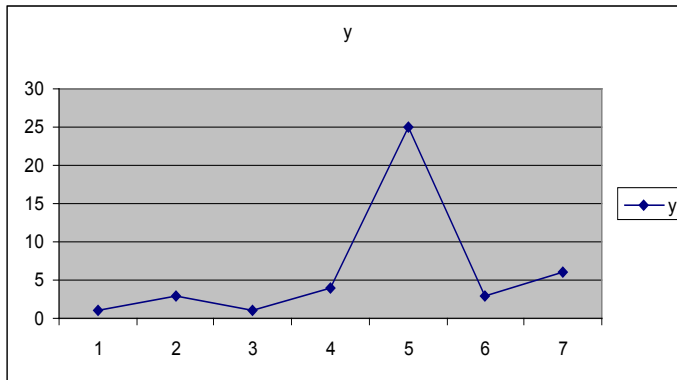
- If the values fluctuate so rapidly, we can do smoothing.
- Smoothing within a window using a moving average
- For example, for window size 3, using the median or mean to smooth
- i.e. mean or median of 3 consecutive values

Smoothing within a Window

Example:

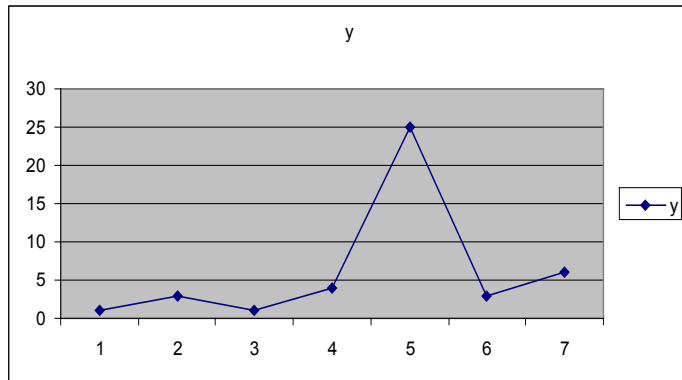
X: [0,1,2,3,4,5,6]

Y: [1,3,1,4,25,6]



Smoothing within a Window

X: [0,1,2,3,4,5,6]
Y: [1,3,1,4,25,6]
smoothed Y : [1.67, 2.67, 10, 10.67, 11.33, 6]



Normalization

- Map the values $x_1, x_2, \dots, x_n$ of the attribute A to a new value x'_i in the interval $[0,1]$ (or any other interval).
- **Min-Max normalization:**

$$x'_i = \frac{x_i - \min_j x_j}{\max_j x_j - \min_j x_j}$$

PRO: This makes the values invariant to rigid displacement of coordinates.

CON: It will encounter an “out-of-bounds” error if a future input case for normalization falls outside of the original data range for A

- **Subtract the mean:** $x'_i = (x_i - \bar{A})$

Z-Score Normalization or Standardization

Z-score normalization:

Scale by mean and standard deviation

$$x'_i = (x_i - \bar{A}) / \sigma_A \quad (1)$$

- Also called: **standard score**, **standardization**
- Positive means value is above the mean, Negative means it is below the mean.
- Data is modified so that it now has the aggregate properties of a standard normal distribution $\mu = 0$ and $\sigma = 1$
- Many analysis and machine learning algorithms benefit from normalizing your data (decision tree methods are the main exception)

Pros and Cons of Standardization

- *Pro:* This method of normalization is useful when the actual minimum and maximum of attribute A are unknown, or when there are outliers that dominate the min-max normalization
- Could be a good thing if you don't mind similar points treated the same (grades)
- *Con:* Normalization may or may not be desirable in some cases. It may make samples that are dispersed in space closer to each other and hence are difficult to separate.
- This could be or unacceptable when small differences matter - e.g. safety-critical scenarios, money in millions

Normalization of Matrix Data

- Both subtraction and z-score normalization methods make the diagonal element of R the moment of inertia of the patterns and forces the origin to coincide with the sample mean
- z-score makes all the elements of inertia unity (i.e. magnitude 1).
- *Con:* Normalization may or may not be desirable in some cases. It may make samples that are dispersed in space *closer to each other* and hence not separable.

Normalization Examples

$$A = (-200, 400, 600, 800)$$

$$\text{Min} = -200, \text{Max} = 800, \text{max-min} = 1000$$

Min-Max Normalization

$$X' = (0, 0.6, 0.8, 1)$$

$$\text{mean}' = 0.6, \sigma_{X'} = 0.374$$

Subtracting Mean Normalization

$$\text{mean} = 400$$

$$X'' = (-600, 0, 200, 400)$$

$$\text{mean}'' = 0, \sigma_{X''} = 100\sqrt{14}$$

Z-score normalization

$$X''' = \left(\frac{-6}{\sqrt{14}}, 0, \frac{2}{\sqrt{14}}, \frac{4}{\sqrt{14}} \right)$$

$$\text{mean}''' = 0, \sigma_{X'''} = 1$$

A COVID Example

- The number of cases in a region, or country, scales by population...
- ...but also other factors.
- How do we compare what a *significant* increase is?
- We need to **normalize** for the range in that region/country in an intelligent way.
- What other examples could there be?

How to compare different sized regions? <https://www.ctvnews.ca/health/coronavirus/covid-19-in-the-u-s-how-do-canada-s-provinces-rank-against-american-states-1.5051033>

Pricing Example

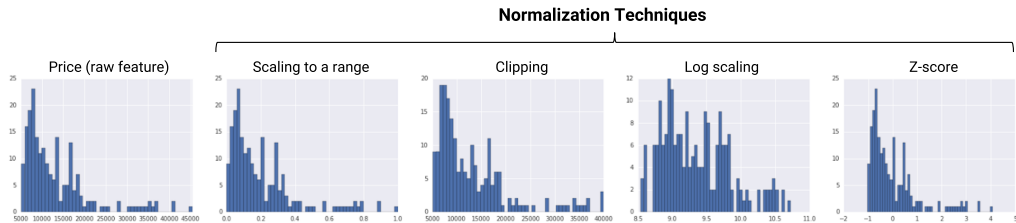


Figure: The following charts show the effect of each normalization technique on the distribution of the raw feature (price) on the left. The charts are based on the data set from 1985 Ward's Automotive Yearbook that is part of the UCI Machine Learning Repository under Automobile Data Set.

From Google Developers Crash Course

(<https://developers.google.com/machine-learning/data-prep/transform/normalization>)

Normalization By Data Scaling

$$x' = \frac{x}{10^j}$$

- Where j is the smallest integer such that $\max |x'| < 1$
- Example:
 - if $x \in [-986, 917]$ then $\max |x| = 986$ then $x' = 1000$
 - So -986 will normalize to -0.986 and 917 to .917
- **Note:** normalization can change the characteristics of the original data. But it is good for comparing values of different scales and reduces influence large numbers in the data.

Normalization of Matrix Data

- If A is the sample-feature matrix in terms of normalized data then let

$$R = \frac{1}{n} A^T A$$
$$r = \frac{1}{n} \sum_{k=1}^n x_{ki} x_{kj}$$

- Under subtraction normalization, R is a covariance matrix
- Under z-score normalization r_{ij} becomes the correlation coefficient between features i and j and $r_{jj} = 1$ for all j
- R is then called the correlation matrix.

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Data Reduction

- **Goal:** improve performance, without hurting accuracy too much
- Numerosity Reduction
 - regression - replace many points with smaller number of predictions or function
 - clustering - much more on time on this later
 - sampling - to reduce data size
- Dimensionality Reduction
 - wavelet transforms
 - principle component analysis (more on this later)

Sampling for Data Reduction

- **Sampling:** obtaining a small subset of datapoints s to represent the whole data set n
- Allows a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Key principle: Choose a representative subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
 - Use adaptive sampling methods such as stratified sampling

Types of Sampling

- **Simple random sampling:** Draw a random number from the sample indices and select the object. Treats all sample as equally likely to be selected.
- **Sampling without replacement:** Remove the objects you select from the remaining samples. Original sample gets reduced every time you make a selection.
- **Sampling with replacement:** A selected object is put back in the original sample. You may select the same object more than one time.
- **Stratified sampling:** Similar to binning where the data set is partitioned and samples are selected from each partition. Good for skewed data.

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Measuring Similarity

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- 1 Definitions and Matrix Measures
- 2 Data Similarity For Binary
 - Contingency Tables
- 3 Data Similarity for Nominal Types
- 4 For Interval, Ratio (and Ordinal) Types
- 5 For Ordinal Types
 - Cosine Similarity

Training and Validation

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Lecture Outline

1 Dividing the Data

- Error Estimation and Training Methods
- Validation

2 Ablation Studies

- Definition
- Uses
- An Example